

Diffie-Hellman Key Agreement

Jordan	Professor Webber
Public Keys = P, G	Public Keys = P, G
Private Key = a	Public Key = b
Key generated= $x = G^a \text{ mod } P$	Key generated= $y = G^b \text{ mod } P$
Exchange of Keys= y	
Secret key generated= $k_a = y^a \text{ mod } P$	Secret key generated= $k_b = x^b \text{ mod } P$
$k^a = k^b$	

Let's say that the modulus (p) is 5 and the base (g) is 12.

Once we have agreed on this, we both choose a secret key (a) for me and (b) for you.

$$a = 3$$

$$b = 6$$

I perform the calculation to give me the number that I will send to you.

$$A = g^a \text{ mod } p$$

Plug in the numbers we agreed upon.

$$12^3 \text{ mod } 5$$

$$1728 \text{ mod } 5$$

$$A = 3$$

You do the same calculation.

$$B = g^b \bmod p$$

$$B = 12^6 \bmod 5$$

$$2985984 \bmod 5$$

$$B = 4$$

I send my result (A) to you, while you send your result (B) to me. We then both calculate the shared secret (s) using the number we received (A) & (B) using this formula:

Jordans calculation:

$$s = B^a \bmod p$$

$$s = 4^3 \bmod 5$$

$$64 \bmod 5$$

$$s = 4$$

Professor Webbers calculation:

$$s = A^b \bmod p$$

$$s = 3^6 \bmod 5$$

$$729 \bmod 5$$

$$s = 4$$

Both of us ended up with the same number for (s), 4. This is our shared secret which only we should know. We can both use this to set up a symmetric encryption allowing us to communicate privately.

First, run Command Prompt as administrator and generate the public parameters

```
openssl genpkey -genparam -algorithm DH -out dhp.pem
```

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.22621.1992]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32> openssl genkey -genparam -algorithm DH -out dhp.pem
```

More dhp.pem:

```
C:\Windows\System32>more dh.pem
-----BEGIN DH PARAMETERS-----
MIIBDAKCAQEA9FFPQqjm0K9PxMrnQqFi+C5Qp2v1jvc8p73KtLm6GmEY+zH8qZ8O
c00gxVqpWpAJ7g7T7Nc7DQCF0o0e/5+6XtHuX0o2gg+HfKVPMT1Tj6/HljevQiiF
QblEDopiO9/ovgq29aK7fB8qiMHKzNh/1Smgnq1ar8vscxQuJxsVL/yoctrDgzIn
U00Bbqlj120ILXeDoolw+6H0E7kz6rMD8wkN00U+jTKDguWphTQ6mCj3LM2v4hhSn
CHais06XXJgnNVyhc4iBsGFJGxlH0KHKiXUSPgZj19aEQjrh1lY+UVWA78R/S/HK
tWPL3lIX37MEgTPX72/lvqPH8IOhC4SwXwIBAgICA0E=
-----END DH PARAMETERS-----
```

To view in plain text, use the following command:

```
openssl pkeyparam -in dhp.pem -text
```

```
C:\Windows\System32>openssl pkeyparam -in dhp.pem -text
-----BEGIN DH PARAMETERS-----
MIIBDAKCAQEAA9FFP0qjm0K9PxMrnQqFi+C5Qp2v1jvc8p73KtLm6GmEY+zH8qZ8O
c0OgxVqpWpA7g7T7Nc7DQCF0oOe/5+6XtHuX0o2gg+HfKVPMT1Tj6/HljevQiiF
QblEDopi09/ovgq29aK7fB8qiMHKzNh/lSmgnq1ar8vscxQuJxsVL/yoctrDgzIn
U00Bqqlj120ILXeDooW+6H0E7kz6rMD8wkN00U+jTKDguWphTQ6mCj3LM2v4hhSn
CHais06XXJgnNVyhc4iBsGFJGx1H0KHKiXUSPgZjl9aEQjrh11Y+UVWA78R/S/HK
tWPL3lIX37MEgTPX72/lvqPH8IOhC4SwXwIBAgICA0E=
-----END DH PARAMETERS-----
DH Parameters: (2048 bit)
P:
00:f4:51:4f:3a:a8:e6:d0:af:4f:c4:ca:e7:42:a1:
62:f8:2e:50:a7:6b:f5:8e:f7:3c:a7:bd:ca:b4:b9:
ba:1a:61:18:fb:31:fc:a9:9f:0e:73:43:a0:c5:5a:
a9:5a:90:09:ee:0e:d3:ec:d7:3b:0d:00:85:d2:83:
9e:ff:9f:ba:5e:d1:ee:5c:ea:36:82:0f:87:7c:a5:
4f:31:3d:53:8f:af:c7:96:37:af:42:28:85:41:b9:
44:0e:8a:62:3b:df:e8:be:0a:b6:f5:a2:bb:7c:1f:
2a:88:c1:ca:cc:d8:7f:95:29:a0:9e:ad:5a:af:cb:
ec:73:14:2e:27:1b:15:2f:fc:a8:72:da:c3:83:32:
27:50:ed:01:aa:a9:63:d7:6d:08:2d:77:83:a2:85:
be:e8:7d:04:ee:4c:fa:ac:c0:fc:c2:43:4e:39:4f:
a3:4c:a0:e0:b9:6a:61:4d:0e:a6:0a:3d:cb:33:6b:
f8:86:14:a7:08:76:a2:b3:4e:97:5c:98:27:35:5c:
a1:73:88:81:b0:61:49:1b:19:47:d0:a1:ca:89:75:
12:3e:06:63:97:d6:84:42:3a:e1:d6:56:3e:51:55:
80:ef:c4:7f:4b:f1:ca:b5:63:cb:de:52:17:df:b3:
04:81:33:d7:ef:6f:e5:be:a3:c7:f0:83:a1:0b:84:
b0:5f
G: 2 (0x2)
```

Each user uses their public parameters to generate their own private and public key saving them in the file dhkey1.pem for Jordan and dhkey2.pem for Professor Webber.

openssl genpkey -paramfile dhp.pem -out dhkey1.pem

openssl pkey -in dhkey1.pem -text -noout

```
C:\Windows\System32>openssl genpkey -paramfile dhp.pem -out dhkey1.pem

C:\Windows\System32>openssl pkey -in dhkey1.pem -text -noout
DH Private-Key: (2048 bit)
private-key:
  44:5c:6e:c5:56:cc:b2:09:30:96:8f:4d:67:c5:a4:
  51:24:a1:ac:0f:63:5e:b9:47:c9:8b:be:ce:99:b2:
  dd:7e:f2:ca:48:9d:28:4c:0c:65:e6:ff:89:2d:cf:
  6c:9d:df:07:94:0c:a5:45:5d:aa:d7:7b:45:21:68:
  6e:44:6b:04:68:c4:de:dc:95:b4:ea:48:f4:83:c7:
  17:19:3a:d4:a5:e0:3f:02:63:5b:2c:4f:82:0a:38:
  f3:de:8c:01:47:ab:6b:89:87:45:35:31:9a:5c:05:
  17:83:e2:5d:84:73:96:66:bc:83:45:ac:f7:11:28:
  b0:58:d9:1f:2f:7b:8c:14:a4:21:74:cb:f4:dd:ee:
  ee:84:5e:e9:5a:37:ff:ef:80:95:25:5a:0f:10:e9:
  5d:7a:16:d2:3c:24:f7:1d:11:1a:de:f5:16:e6:32:
  1b:f9:ff:d0:55:31:80:84:16:6f:c3:fb:cd:f3:f6:
  a6:0e:cd:9c:df:7b:c1:a2:64:9c:17:15:78:20:21:
  77:d9:46:cd:45:e9:15:d6:6b:6e:21:54:a5:72:8b:
  64:cb:af:e5:3b:1d:dd:a8:68:dd:2e:21:c8:ef:16:
  ba:6a:5b:18:34:a8:b8:4e:41:b5:41:fe:8f:0c:0b:
  7f:44:48:94:17:3b:de:88:64:db:82:e2:23:9b:59:
  28
public-key:
  56:0e:22:62:f0:fa:d2:07:d5:67:c3:ab:35:e9:13:
  0f:fa:c0:de:39:7a:7f:4d:a1:ed:92:5c:69:ba:fd:
  b3:d8:de:4a:ff:1a:32:02:25:04:41:99:87:7e:f3:
  11:49:55:2d:58:9e:05:90:d7:b4:76:a4:b7:60:ce:
  72:36:43:fb:bb:d3:f8:16:51:75:6e:90:41:09:48:
  ac:53:81:76:35:ea:52:69:8b:63:99:c3:e4:ae:fd:
  24:0a:cc:e2:48:e3:ae:9c:6b:c7:43:60:56:62:99:
  8b:56:59:1d:06:2f:9e:38:d4:02:44:53:48:ef:d6:
  07:77:07:46:ef:4c:2f:4e:43:98:99:2e:9c:7e:65:
  64:71:38:e5:96:f9:ff:cf:92:1e:83:b3:5a:1b:17:
  c3:8c:dd:37:52:1d:46:63:25:9e:8e:23:dd:f6:2b:
  f8:78:67:8a:8f:af:51:5a:69:7d:1d:b1:4c:7f:c4:
  11:58:61:94:b3:63:f5:14:3b:03:08:b0:fe:be:d4:
  7f:45:d2:d8:45:e9:f5:88:f7:a7:d1:54:19:d2:0d:
  d2:5b:72:54:d6:35:ff:76:b1:2c:8a:57:48:12:0e:
  0b:86:cb:fc:eb:e8:ca:1f:d0:48:32:5c:f8:db:f9:
  0c:93:34:c4:ea:bd:a7:24:0f:2b:68:01:49:51:f2:
  c3
P:
  00:f4:51:4f:3a:a8:e6:d0:af:4f:c4:ca:e7:42:a1:
  62:f8:2e:50:a7:6b:f5:8e:f7:3c:a7:bd:ca:b4:b9:
  ba:1a:61:18:fb:31:fc:a9:9f:0e:73:43:a0:c5:5a:
  a9:5a:90:09:ee:0e:d3:ec:d7:3b:0d:00:85:d2:83:
  9e:ff:9f:ba:5e:d1:ee:5c:ea:36:82:0f:87:7c:a5:
  4f:31:3d:53:8f:af:c7:96:37:af:42:28:85:41:b9:
  44:0e:8a:62:3b:df:e8:be:0a:b6:f5:a2:bb:7c:1f:
  2a:88:c1:ca:cc:d8:7f:95:29:a0:9e:ad:5a:af:cb:
  ec:73:14:2e:27:1b:15:2f:fc:a8:72:da:c3:83:32:
  27:50:ed:01:aa:a9:63:d7:6d:08:2d:77:83:a2:85:
  be:e8:7d:04:ee:4c:fa:ac:c0:fc:c2:43:4e:39:4f:
  a3:4c:a0:e0:b9:6a:61:4d:0e:a6:0a:3d:cb:33:6b:
  f8:86:14:a7:08:76:a2:b3:4e:97:5c:98:27:35:5c:
  a1:73:88:81:b0:61:49:1b:19:47:d0:a1:ca:89:75:
  12:3e:06:63:97:d6:84:42:3a:e1:d6:56:3e:51:55:
  80:ef:c4:7f:4b:f1:ca:b5:63:cb:de:52:17:df:b3:
  04:81:33:d7:ef:6f:e5:be:a3:c7:f0:83:a1:0b:84:
  b0:5f
G:      2 (0x2)

C:\Windows\System32>
```

You use the same parameters, dhkey.pem, to generate your private/public key:

```
openssl genpkey -paramfile dhkey.pem -out dhkey2.pem
```

Use openssl pkey -in dhkey2.pem -text -noout to view in plain text:

```
C:\Windows\System32> openssl genpkey -paramfile dhkey.pem -out dhkey2.pem

C:\Windows\System32>openssl pkey -in dhkey2.pem -text -noout
DH Private-Key: (2048 bit)
private-key:
 41:fe:a1:b5:8a:f2:c6:db:e0:e8:8d:f1:80:5b:f7:
 03:bc:d4:23:83:ba:15:f8:97:2f:8b:c9:fe:c6:5c:
 77:40:79:24:55:ad:d4:bd:00:1c:cc:43:e6:4c:df:
 18:96:94:09:4f:8c:9c:78:e7:35:ac:14:5d:94:40:
 b7:b3:5f:e6:da:f2:41:c1:22:2b:03:4d:53:55:96:
 3e:90:67:fd:9b:23:78:13:75:8b:ed:40:56:c0:35:
 00:de:0b:2c:d3:74:0c:2a:14:ad:11:e3:2a:f3:b8:
 76:61:8e:93:4e:1b:ec:ab:6b:d5:dc:8e:20:cd:3c:
 0f:b1:8a:0b:78:e5:bd:09:72:e0:bb:13:65:0e:41:
 91:f0:88:e3:e3:60:c4:3c:50:f5:a6:a5:c2:8b:43:
 41:5c:92:71:87:54:37:6e:3d:48:40:2e:8d:36:bd:
 f1:9f:40:84:17:d5:67:cb:f6:ad:0f:8f:52:10:16:
 af:c5:ce:78:fb:54:c8:db:4a:1e:6c:df:52:ab:3d:
 01:56:bd:c3:ab:07:e1:81:7a:df:6c:8b:43:6c:fb:
 aa:55:53:66:3a:48:b9:05:09:db:e3:44:69:6b:83:
 2f:46:b1:88:db:33:3c:4a:9d:50:0c:38:a0:b6:75:
 da:9b:f6:3a:ff:10:6c:1e:5e:47:98:7e:d0:ff:1d:
 c3
public-key:
 00:f4:47:fb:6e:fb:b9:40:71:76:64:14:09:6f:f3:
 ed:25:32:d4:53:5e:07:1e:9a:6c:76:05:48:64:2c:
 6c:f4:b0:41:ad:d6:ad:08:c0:4e:7f:f3:3f:4a:26:
 7d:7b:d8:6e:b8:f6:11:f6:4b:55:a5:25:68:4b:72:
 c1:13:66:fb:72:7f:85:bf:03:5d:9d:2a:97:5d:ab:
 4a:da:a1:d6:19:7b:fb:5e:1a:da:bf:25:69:b0:d6:
 dd:fa:a6:50:ae:96:6a:7a:e6:a5:58:52:e4:1c:9f:
 71:7e:e7:cc:39:ca:b4:94:ad:19:16:23:38:6a:60:
 55:62:7c:4c:7d:1f:c4:79:88:02:05:95:8f:4c:86:
 46:76:67:4a:fb:9b:e5:7d:85:2f:0b:f3:f7:a1:bb:
 17:ef:13:60:ec:9c:49:dc:25:c2:74:e5:82:a0:2e:
 6b:54:c3:4c:58:c2:24:a1:ee:07:07:1f:8d:b0:e7:
 10:6e:ef:7a:96:d3:d6:c4:92:6d:f9:62:c7:90:82:
 2e:12:af:66:aa:55:87:e8:8e:6c:f1:c8:cf:54:38:
 0a:92:cd:0b:5c:ff:00:2b:d6:d4:19:69:7f:ac:7f:
 f0:89:35:a6:2d:29:a6:5b:b8:bc:50:98:d1:0a:cf:
 eb:fe:f3:d6:6c:e8:5a:55:e2:4c:a0:1f:ef:fb:db:
 6c:3e
P:
 00:f4:51:4f:3a:a8:e6:d0:af:4f:c4:ca:e7:42:a1:
 62:f8:2e:50:a7:6b:f5:8e:f7:3c:a7:bd:ca:b4:b9:
 ba:1a:61:18:fb:31:fc:a9:9f:0e:73:43:a0:c5:5a:
 a9:5a:90:09:ee:0e:d3:ec:d7:3b:0d:00:85:d2:83:
 9e:ff:9f:ba:5e:d1:ee:5c:ea:36:82:0f:87:7c:a5:
 4f:31:3d:53:8f:af:c7:96:37:af:42:28:85:41:b9:
 44:0e:8a:62:3b:df:e8:be:0a:b6:f5:a2:bb:7c:1f:
 2a:88:c1:ca:cc:d8:7f:95:29:a0:9e:ad:5a:af:cb:
 ec:73:14:2e:27:1b:15:2f:fc:a8:72:da:c3:83:32:
 27:50:ed:01:aa:a9:63:d7:6d:08:2d:77:83:a2:85:
 be:e8:7d:04:ee:4c:fa:ac:c0:fc:c2:43:4e:39:4f:
 a3:4c:a0:e0:b9:6a:61:4d:0e:a6:0a:3d:cb:33:6b:
 f8:86:14:a7:08:76:a2:b3:4e:97:5c:98:27:35:5c:
 a1:73:88:81:b0:61:49:1b:19:47:d0:a1:ca:89:75:
 12:3e:06:63:97:d6:84:42:3a:e1:d6:56:3e:51:55:
 80:ef:c4:7f:4b:f1:ca:b5:63:cb:de:52:17:df:b3:
 04:81:33:d7:ef:6f:e5:be:a3:c7:f0:83:a1:0b:84:
 b0:5f
G: 2 (0x2)
```

We both must exchange our public keys. First, we extract the public key into the file

dhpub(1).pem and for you dhpub(2).pem:

openssl pkey -in dhkey1.pem -pubout -out dhpub1.pem

openssl pkey -in dhkey2.pem -pubout -out dhpub2.pem

To view this in plain text use these commands:

openssl pkey -pubin -in dhpub1.pem -text

openssl pkey -pubin -in dhpub2.pem -text

```
C:\Windows\System32>openssl pkey -pubin -in dhpub1.pem -text
```

```
-----BEGIN PUBLIC KEY-----
```

```
MIICJCCARCGCSqGSIb3DQEDATCCAQgCggEBApRRtZqo5tCvT8TK50KhYvVguUKdr
9Y73PKe9yrS5UhpGPsx/KmFDnDoMvQVqQCe400+zX0w0AhdKDNv+fu17R7lZq
NoIPh3y1TzE9U4+vx5Y3r0IohUG5RA6KYjvF6L4KtVWiu3wFkojByszYf5UpoJ6t
Wq/L7HMULicbFS/8qHLaw4MyJ1DtAaqP9dTC13g6KFvuh9B05M+qzA/MJDTj1P
o0yg4LlqYU00pgo9yzNr+IYUpwh2orNO11yYJzVcoX0IgbBhSRsZR9Chyo11Ej4G
Y5fWheI64dZWPlFVgO/Ef0vxYrVjy95SF9+zBIEz1+9v5b6jx/CD0QuEsF8CAQID
ggEFAAKCAQBDIj18PrSB9Vnw6s16RMP+sDeOxP/TaHtk1xpUv2z2N5K/xoyAiUE
QZmHfVMSRvUtwJ4FkNe0dQ53YM5YnkP7u9P4F1F1bpBBCUIsU4F2NepSaYtjmcPk
rv0kCsziS00unGvH02BwYpMLV1kd8i+eONQCRFNI79YHdwG70wvTk0YmS6cfmVk
cTjllvn/z5Ieg7NaGxfDJN03Uhi1GYWejiPd9iv4eGeKj69Rwm19HbFMF8QRWGGU
s2P1FDsDCLD+vER/RdLYRen1iPen0VQZ0g35W3J1UjX/drEsildIEg4Lhsv86+jK
H9BIMLz42/kMKzTE6r2nJ4A8raAFJUFld
```

```
-----END PUBLIC KEY-----
```

```
DH Public-Key: (2048 bit)
```

```
public-key:
```

```
56:0e:22:62:f0:fa:d2:07:d5:67:c3:ab:35:e9:13:
0f:fa:c0:de:39:7a:7f:4d:a1:ed:92:5c:69:ba:fd:
b3:d8:de:4a:ff:1a:32:02:25:04:d1:99:87:7e:f3:
11:49:55:2d:58:9e:05:90:d7:b4:76:a4:b7:60:ce:
72:36:43:fb:bb:d3:f8:16:51:75:6e:90:41:09:48:
ac:53:81:76:35:ea:52:69:8b:63:99:c3:e4:ae:fd:
24:0a:cc:e2:48:e3:ae:9c:6b:c7:43:60:56:62:99:
8b:56:59:1d:06:2f:9e:38:d4:02:44:53:48:ef:d6:
07:77:07:46:ef:4c:2f:4e:43:98:99:2e:9c:7e:65:
64:71:38:e5:96:f9:ff:cf:92:1e:83:b3:5a:1b:17:
c3:8c:dd:37:52:1d:46:63:25:9e:8e:23:dd:f6:2b:
f8:78:67:8a:8f:af:51:5a:69:7d:1d:b1:4c:7f:c4:
11:58:61:94:b3:63:f5:14:3b:03:08:b0:fe:be:d4:
7f:45:d2:d8:45:e9:f5:88:f7:a7:d1:54:19:d2:0d:
d2:5b:72:54:d6:35:ff:76:b1:2c:8a:57:48:12:0e:
0b:86:cb:fc:eb:e8:ca:1f:d0:48:32:5c:f8:db:f9:
0c:93:34:c4:ea:bd:a7:24:0f:2b:68:01:49:51:f2:
c3
```

```
P:
```

```
00:f4:51:4f:3a:a8:e6:d0:af:4f:c4:ca:e7:42:a1:
62:f8:2e:50:a7:6b:f5:8e:f7:3c:a7:bd:ca:b4:b9:
ba:1a:61:18:fb:31:fc:a9:9f:0e:73:43:a0:c5:5a:
a9:5a:90:09:ee:0e:d3:ec:d7:3b:0d:00:85:d2:83:
9e:ff:9f:ba:5e:d1:ee:5c:ea:36:82:0f:87:7c:a5:
4f:31:3d:53:8f:af:c7:96:37:af:42:28:85:41:b9:
44:0e:8a:62:3b:df:e8:be:0a:b6:f5:a2:bb:7c:1f:
2a:88:c1:ca:cc:d8:7f:95:29:a0:9e:ad:5a:af:cb:
ec:73:14:2e:27:1b:15:2f:fc:a8:72:da:c3:83:32:
27:50:ed:01:aa:a9:63:d7:6d:08:2d:77:83:a2:85:
be:e8:7d:04:ee:4c:fa:ac:c0:fc:c2:43:4e:39:4f:
a3:4c:a0:e0:b9:6a:61:4d:0e:a6:0a:3d:cb:33:6b:
f8:86:14:a7:08:76:a2:b3:4e:97:5c:98:27:35:5c:
a1:73:88:81:b0:61:49:1b:19:47:d0:a1:ca:89:75:
12:3e:06:63:97:d6:84:42:3a:e1:d6:56:3e:51:55:
80:ef:c4:7f:4b:f1:ca:b5:63:cb:de:52:17:df:b3:
04:81:33:d7:ef:6f:e5:be:a3:c7:f0:83:a1:0b:84:
b0:5f
```

```
G: 2 (0x2)
```

```
C:\Windows\System32>openssl pkey -pubin -in dhpub2.pem -text
```

```
-----BEGIN PUBLIC KEY-----
```

```
MIICJCCARCGCSqGSIb3DQEDATCCAQgCggEBApRRtZqo5tCvT8TK50KhYvVguUKdr
9Y73PKe9yrS5UhpGPsx/KmFDnDoMvQVqQCe400+zX0w0AhdKDNv+fu17R7lZq
NoIPh3y1TzE9U4+vx5Y3r0IohUG5RA6KYjvF6L4KtVWiu3wFkojByszYf5UpoJ6t
Wq/L7HMULicbFS/8qHLaw4MyJ1DtAaqP9dTC13g6KFvuh9B05M+qzA/MJDTj1P
o0yg4LlqYU00pgo9yzNr+IYUpwh2orNO11yYJzVcoX0IgbBhSRsZR9Chyo11Ej4G
Y5fWheI64dZWPlFVgO/Ef0vxYrVjy95SF9+zBIEz1+9v5b6jx/CD0QuEsF8CAQID
ggEFAAKCAQEA9Ef7bvU5QHf2ZBQJb/PtJTLUU14HppdsdVIZCxs9LBBrdatCMBO
f/M/SiZ9e9huuPYR9ktVpSVoS3LBE2b7cn+FvWnDnSgXXatK2qHMGXv7XhravyVp
sNbd+qZQrpZqeualWFLKHJ39xfufMocq0Lk0ZFIM4amBVYnxMfR/EeYgC8ZWPTIZG
dmdK+5vIfYUvC/P3obsX7xNg7JxJ3CXCDOWCoC5rVMNMWMIkoe4HBx+N5OcQbu96
1tPwxJ3t+WLHKIIuEq9mq1WH6IS8cjpVDgKks0LXP8AK9bUGW1/rH/wlTWmLsmm
W7i8UJjRCs/r/vPwbOhaVeJMo8/v+9tsgP==
```

```
-----END PUBLIC KEY-----
```

```
DH Public-Key: (2048 bit)
```

```
public-key:
```

```
00:f4:47:fb:6e:fb:b9:40:71:76:64:14:09:6f:f3:
ed:25:32:d4:53:5e:07:1e:9a:6c:76:05:48:64:2c:
6c:f4:b0:41:ad:d6:ad:08:c0:4e:7f:f3:3f:4a:26:
7d:7b:d8:6e:b8:f6:11:f6:4b:55:a5:25:68:4b:72:
c1:13:66:fb:72:7f:85:bf:03:5d:9d:2a:97:5d:ab:
4a:da:a1:d6:19:7b:fb:5e:1a:da:bf:25:69:b0:d6:
dd:fa:a6:50:ae:96:6a:7a:e6:a5:58:52:e4:1c:9f:
71:7e:e7:cc:39:ca:b4:94:ad:19:16:23:38:6a:60:
55:62:7c:4c:7d:1f:c4:79:88:02:05:95:8f:4c:86:
46:76:67:4a:fb:9b:e5:7d:85:2f:0b:f3:f7:a1:bb:
17:ef:13:60:ec:9c:49:dc:25:c2:74:e5:82:a0:2e:
6b:54:c3:4c:58:c2:24:a1:ee:07:07:1f:8d:b0:e7:
10:6e:ef:7a:96:d3:d6:c4:92:6d:f9:62:c7:90:82:
2e:12:af:66:aa:55:87:e8:8e:6c:f1:c8:cf:54:38:
0a:92:cd:0b:5c:ff:00:2b:d6:d4:19:69:7f:ac:7f:
f0:89:35:a6:2d:29:a6:5b:b8:bc:50:98:d1:0a:cf:
eb:fe:f3:d6:6c:e8:5a:55:e2:4c:a0:1f:ef:fb:db:
6c:3e
```

```
P:
```

```
00:f4:51:4f:3a:a8:e6:d0:af:4f:c4:ca:e7:42:a1:
62:f8:2e:50:a7:6b:f5:8e:f7:3c:a7:bd:ca:b4:b9:
ba:1a:61:18:fb:31:fc:a9:9f:0e:73:43:a0:c5:5a:
a9:5a:90:09:ee:0e:d3:ec:d7:3b:0d:00:85:d2:83:
9e:ff:9f:ba:5e:d1:ee:5c:ea:36:82:0f:87:7c:a5:
4f:31:3d:53:8f:af:c7:96:37:af:42:28:85:41:b9:
44:0e:8a:62:3b:df:e8:be:0a:b6:f5:a2:bb:7c:1f:
2a:88:c1:ca:cc:d8:7f:95:29:a0:9e:ad:5a:af:cb:
ec:73:14:2e:27:1b:15:2f:fc:a8:72:da:c3:83:32:
27:50:ed:01:aa:a9:63:d7:6d:08:2d:77:83:a2:85:
be:e8:7d:04:ee:4c:fa:ac:c0:fc:c2:43:4e:39:4f:
a3:4c:a0:e0:b9:6a:61:4d:0e:a6:0a:3d:cb:33:6b:
f8:86:14:a7:08:76:a2:b3:4e:97:5c:98:27:35:5c:
a1:73:88:81:b0:61:49:1b:19:47:d0:a1:ca:89:75:
12:3e:06:63:97:d6:84:42:3a:e1:d6:56:3e:51:55:
80:ef:c4:7f:4b:f1:ca:b5:63:cb:de:52:17:df:b3:
04:81:33:d7:ef:6f:e5:be:a3:c7:f0:83:a1:0b:84:
b0:5f
```

```
G: 2 (0x2)
```


After we exchanged our public keys each of us can proceed to generate our secret key.

Using the following command: `openssl pkeyutl -derive -inkey dhkey1.pem -peerkey dhp2b2.pem -out secret1.bin`

`openssl pkeyutl -derive -inkey dhkey2.pem -peerkey dhp1b1.pem -out secret2.bin`

On windows you can compare the files using the command: `comp.exe secret1.bin secret2.bin`.

The secrets should be the same as we determined in the formulas above.

To view them you need to use a hexdump or hex editor I used Visual Studio Code's Hex editor:

Secret 1:

```
secret1.zip X secret2.zip
C: > Users > Chron > OneDrive > Desktop > secret1.zip
00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F Decoded Text
00000000 50 4B 03 04 14 00 00 00 00 00 27 65 FC 56 90 23 P K . . . . . ' e . V . #
00000010 3B 82 00 01 00 00 00 01 00 00 0B 00 00 00 73 65 ; . . . . . s e
00000020 63 72 65 74 31 2E 62 69 6E 77 AE D9 CF 2E 28 E0 c r e t 1 . b i n w . . . ( .
00000030 B5 EB CA 2F 9A B6 C8 93 B2 20 99 A2 5D 22 8C F2 . . . / . . . . . ] " . .
00000040 A8 DB A7 B9 65 92 92 04 89 23 81 5A EF 18 95 CB . . . e . . . . # . Z . . .
00000050 44 82 74 8E 2D 5B 87 00 53 72 86 8E C1 25 3E 6B D . t . - [ . . S r . . % > k
00000060 45 E0 D1 B7 3C 8B E4 23 E8 7B B8 EE 02 2C 7B D1 E . . . < . . # . { . . . , { .
00000070 D4 EB 70 80 69 18 4C 04 50 F4 EE B1 E3 21 A8 27 . . p . i . L . P . . . . ! . '
00000080 DF 95 7E A4 D3 D6 DE AA 47 73 E2 7A 94 BE 8D 29 . . ~ . . . . . G s . z . . . )
00000090 59 67 83 D0 AB 14 84 18 A2 86 51 7E 66 77 01 59 Y g . . . . . Q ~ f w . Y
000000A0 4E 29 7A 0B 00 9E C6 E1 5D AC 0B 89 64 73 0D F7 N ) z . . . . . ] . . . d s . .
000000B0 64 F8 72 00 57 0A 4E E7 10 59 B7 C2 37 38 F7 30 d . r . W . N . . Y . . 7 8 . 0
000000C0 3C 48 15 A0 B6 E5 65 E3 2A 91 59 B0 A5 F2 99 93 < H . . . . e . * . Y . . . .
000000D0 69 87 C9 99 3B 60 11 7A ED 9D EE DA 16 6B C4 36 i . . . ; ` . z . . . . . k . 6
000000E0 4B F1 62 CA F7 FE CE E4 11 D9 8C 21 18 40 C2 76 K . b . . . . . ! . @ . v
000000F0 23 02 A8 C0 6B A4 60 62 47 9B F6 91 1F C2 39 82 # . . . k . ` b G . . . . . 9 .
00000100 4F CB 2B BE 34 CE 5A FF BD C8 A6 E2 B2 72 3A 71 O . + . 4 . Z . . . . . r : q
00000110 FE 51 10 4D D1 19 17 6F D9 BF 6F 15 74 71 E1 F4 . Q . M . . . o . . o . t q . .
00000120 3C DF A3 34 A3 D2 A9 15 A7 50 4B 01 02 14 00 14 < . . 4 . . . . . P K . . . .
00000130 00 00 00 00 00 27 65 FC 56 90 23 3B 82 00 01 00 . . . . . ' e . V . # ; . . . .
00000140 00 00 01 00 00 0B 00 00 00 00 00 00 00 00 20 . . . . .
00000150 00 00 00 00 00 00 00 73 65 63 72 65 74 31 2E 62 . . . . . s e c r e t 1 . b
00000160 69 6E 50 4B 05 06 00 00 00 00 01 00 01 00 39 00 i n P K . . . . . 9 .
00000170 00 00 29 01 00 00 00 00 . . . ) . . . .
```

Secret 2:

secret1.zipsecret2.zip X

C: > Users > Chron > OneDrive > Desktop > secret2.zip

🌀

00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F

Decoded Text

PK - e . V . #
; s e
c r e t 2 . b i n w (.
. . . /] " . .
. . . e # . Z
D . t . - [. . S r . . . % > k
E . . . < . . # . { . . . , { .
. . p . i . L . P ! . '
. . ~ G s . z . . .)
Y g Q ~ f w . Y
N) z] . . . d s . .
d . r . W . N . . Y . . 7 8 . 0
< H e . * . Y
i . . . ; ` . z k . 6
K . b ! . @ . v
. . . k . ` b G 9 .
O . + . 4 . Z r : q
. Q . M . . . o . . o . t q . .
< . . 4 P K
. - e . V . # ;
.
. s e c r e t 2 . b
i n P K 9 .
. .)